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ARTICLE

Distribution of platinum-group elements in magmatic and altered ores in the Jinchuan intrusion, China: an example of selenium remobilization by postmagmatic fluids

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Abstract The division of platinum-group elements (PGE) between those hosted in platinum-group minerals (PGM) versus those in solid solution in base metal sulfides (BMS) has been determined for ores from the PGE-bearing Ni-Cu-rich Jinchuan intrusion in northwest China. All the BMS are devoid of Pt and Ir, and magmatic BMS are also barren of Rh. These PGE may have been scavenged by arsenic to form PGM during magmatic crystallization of the BMS. Pd, Os, and Ru are recorded in BMS and Pd is predominantly in solid solution in pentlandite. Unlike the fresh magmatic ores, in altered or serpentinized ores, Pd-PGM are present. Froodite is hosted in magnetite, formed during alteration of BMS, accompanied by sulfur loss and liberation of Pd. Michenerite ([Pd, Pt]BiTe), sperrylite ($PtAs_2$), and Au-bearing PGM are located in altered silicates. Irarsite (IrAsS) occurs mainly enclosed in BMS. Padmaite (PdBiSe), identified at the junctions of magnetite and BMS, was the last PGM to form and locally partially replaces earlier non-Se-bearing PGM. We propose that padmaite formed under oxidizing conditions during late local remobilization of Se from the BMS. Se-bearing PGM are rare and our review shows they are frequently associated with carbonate, suggesting that Pd and Se can be mobilized great distances in low pH oxidizing fluids and may be precipitated on contact with carbonate. S/Se ratios are used by researchers of magmatic Ni-Cu-PGE ores to determine sulfur loss, assuming Se is immobile and representative of magmatic sulfur content. This study shows that Se as well as S is potentially mobile and this should be considered in the use of S/Se ratios.

Keywords Jinchuan · Platinum · Base metal sulfides · Selenium · Carbonate

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Introduction

Until recently, it has not been easy to establish the distribution of platinum-group elements (PGE) between base metal sulfides (BMS) and platinum-group minerals (PGM). This is due to difficulties in detecting PGE in solid solution in the BMS at only parts per million (ppm) concentrations, usually below the detection limits of an electron microprobe. The history of analytical development of measurement of PGE in BMS has been summarized in Cabri et al. (2010) who state that although the detection of PGE using microprobe techniques has improved, it was necessary to use microparticle-induced X-ray emission (PIXE), then secondary ion mass spectrometry, and more recently laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) to obtain lower levels of detection of the PGE. Using LA-ICP-MS, it is now possible to routinely measure trace element concentrations of PGE in BMS and this has been shown to compare favorably with micro-PIXE results (Cabri et al. 2003). Thus, it is now possible to determine the distribution of PGE between PGM and in solid solution in BMS.

Although the concentrations of PGE in the BMS are much lower than in PGM, the abundance of the BMS is

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much greater and so both may host considerable quantities of PGE. The individual PGE exhibit different behaviors when it comes to their distribution between PGM and BMS. Recent research has shown a common pattern of distribution of PGE between BMS with Os, Ir, Ru, and Rh concentrated in pentlandite and pyrrhotite and Pd concentrated mainly in pentlandite. In contrast, Pt and Au do not partition into BMS and instead form discrete precious metal minerals (e.g., Cabri 1992, Huminicki et al. 2005, 2008, Holwell and McDonald 2007, 2010; Godel et al. 2007; Godel and Barnes 2008a; Barnes et al. 2008).

The initial distribution of PGE in an igneous intrusion is controlled by the magmatic processes. Sulfide liquid may separate from silicate magma and then this sulfide liquid will crystallize and fractionate as it cools. An initial monosulfide solid solution (MSS) crystallizes, exolving to form pyrrhotite and pentlandite, leaving a Cu-rich liquid which then crystallizes to form intermediate solid solution (ISS), exsolving chalcopyrite on cooling (e.g., Naldrett 2004). The Ni-Cu-rich sulfide liquid scavenges PGE (e.g., Barnes and Lightfoot 2005) which may be held in solid solution in the BMS or may crystallize as PGM directly from the sulfide melt (Helmy et al. 2007; Holwell and McDonald 2007; Hutchinson and McDonald 2008; McDonald 2008, Dare et al. 2010a). Both Pt and Pd are likely to partition into Cu-rich liquid post-MSS crystallization and then, in some cases, PGE may also be concentrated into and crystallize from a late semimetal-rich (As-Bi-Te) melt (e.g., Cabri and Laflamme 1976; Tomkins 2010; Godel et al. 2012).

The details of the distribution and redistribution of the PGE after the BMS have crystallized and go through subsolidus reactions are much less well understood. There has been some experimental work on the partitioning of PGE into BMS at magmatic and subsolidus temperatures (e.g., Makovicky 2002). The upper limits of PGE that can be held in solid solution in BMS may differ under varying conditions. Holwell et al. (2011) have measured concentrations of up to 900 ppm Pt and up to 680 ppm Pd in homogenized sulfide melt inclusions from the Platreef deposit in the Bushveld Complex. Barnes et al. (2008) proposed that PGE remain in BMS if cooling is fast but exsolve from BMS if subsolidus cooling is slow. PGE may diffuse subsolidus and this may lead, for example, to pentlandite commonly hosting Pd (e.g., Dare et al. 2010b). Mobility of PGE during alteration at low temperatures is better understood as PGE recrystallize to form PGM during serpentinization (e.g., Prichard et al. 1994; 2001; Seabrook et al. 2004; Wang et al. 2008). The ability to analyze PGM and PGE in BMS allows both igneous and metamorphic processes of PGE concentration to be examined in detail. The evidence for these processes is held within the textures and geochemistry of the PGM and Ni-Cu-sulfide ores (Ballhaus and Sylvester 2000; Barnes et al. 2006; Godel et al. 2007; Holwell and McDonald 2007; Godel and Barnes 2008a, b; Barnes et al. 2008).

Major Ni deposits usually contain PGE-bearing BMS and the Jinchuan igneous intrusion, situated in northwest China, is the third largest Ni deposit in the world after Noril'sk and Sudbury. It has ore reserves of 500 million metric tons at Ni and Cu grades of 1.2 and 0.7 wt%, respectively, with an average PGE + Au grade in the sulfide ores of about 1 ppm (Chai and Naldrett 1992a). This complex consists predominantly of ultramafic igneous rocks and was intruded into marble and gneiss. The Ni-Cu-PGE ores are dominated by net textured BMS with minor disseminated and rarely massive sulfides. These ores have undergone postmagmatic hydrothermal alteration and shear deformation. Sheared fabrics are observed only locally whereas hydrothermal alteration is much more pervasive, although variable as demonstrated by the preservation of unaltered primary magmatic textures and minerals in some parts of the intrusion (Zhou et al. 2002; Li et al. 2004). There is a lack of information on both the distribution of PGE in the BMS ores and the types of PGM that occur in Jinchuan. Padmaite (PdBiSe), sperrylite ($PtAs_2$), and irarsite (IrAsS) have been described by Prichard et al. (2005). Sperrylite, froodite ($PdBi_2$), and michenerite ([Pd,Pt]BiTe) have also been reported in a brief study by Su et al. (2008) with Yang et al. (2006) identifying moncheite ($PtTe_2$), merenskyite ($PdTe_2$), kotulskite (PdTe), and Ag-Pd-Te-Bi compounds in addition to those already reported.

This paper describes the total distribution of PGE between PGM and as traces of PGE in solid solution in BMS in a suite of samples from Jinchuan analyzed and described by Song et al. (2006). Fresher magmatic as well as more altered and deformed ores have been examined. The aim of this study has been to identify the mobility of the PGE into different hosts during both magmatic and later alteration processes and so develop a paragenetic sequence of PGE-bearing mineral formation. The occurrence and origin of rare and unusual Se-bearing PGM located in these ores from Jinchuan are compared with other Se-bearing PGM described elsewhere. Common factors required for the origin of Se-bearing PGM are suggested and as a consequence of this, the suitability of the use of S/Se ratios is discussed.

Geology of Jinchuan

The Jinchuan intrusion is situated at the northern margin of the fault-bounded Longshoushan Belt which occurs along the southwestern margin of the North China Craton (Fig. 1). This belt consists of migmatites, gneisses, and serpentine marbles belonging to the Baijiazuizi complex that overlies schists and banded marbles of the Tamazigou complex; both complexes have been intruded by multiple generations of granitic and mafic dykes. Lesser metamorphosed strata

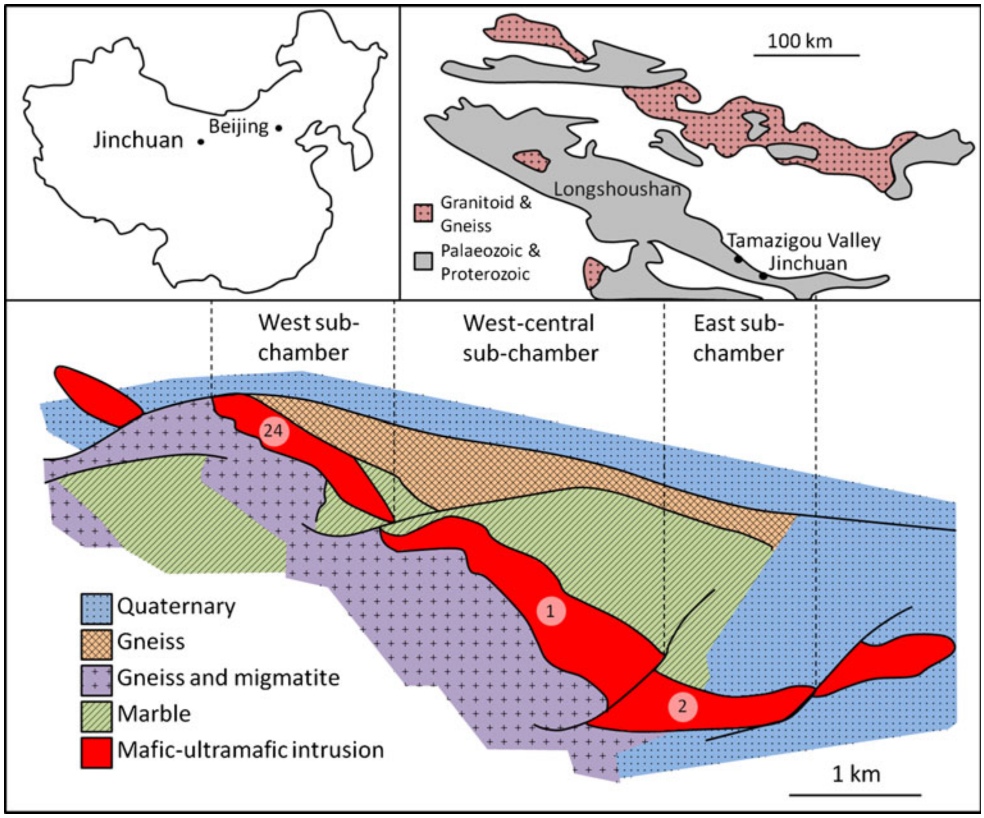

Fig. 1 Location and geological map of the Jinchuan intrusion showing the positions of ore bodies 1, 2, and 24 (redrawn after Chai and Naldrett (1992b) and Lehmann et al. (2007))

consisting mostly of Denzhigou group conglomerate, sandstone, and limestone and the younger Hanmushan sericite–quartz schists, calcareous schists, and limestone breccias unconformably overlie these complexes (Chai and Naldrett 1992b; Zhou et al. 2002).

The Jinchuan intrusion is approximately 6,000 m long, 300 m wide, and more than 1,000 m thick in its central part. It crosscuts the marbles and gneisses of the Baijiazuizi complex and strikes roughly northwest–southeast, parallel to the regional structural trend. The intrusion is now understood to be a highly tilted sill (Lehmann et al. 2007) and is cut by a series of northeast trending strike slip faults dividing the intrusion into sections. Originally, the complex was divided into mining areas III, I, II, and IV by the Sixth Geological Unit of the Geological Survey of Gansu Province (Sixth Geological Unit 1984). However, based on the geometry and distribution of rock types, Chai and Naldrett (1992b) divided the intrusion into three magma subchambers; the west, west-central, and east subchambers which host ore bodies 24, 1, and 2, respectively (Fig. 1).

The intrusion is essentially composed of five rock types in decreasing order of abundance; lherzolite, dunite, plagioclase lherzolite, olivine websterite, and websterite (Chai and Naldrett 1992b, Zhou et al. 2002; Yang et al. 2006). The crystallization sequence of the primary igneous minerals in the order from first to last is chromite, olivine, orthopyroxene, clinopyroxene, and plagioclase (Yang et al. 2006) and the intrusion is considered to be an ultramafic cumulate sequence produced by crystallization of a high Mg basaltic magma (Chai and Naldrett 1992b). Net textured, disseminated, and rare massive ore have been found to be composed principally of primary pyrrhotite, pentlandite, and chalcopyrite as well as secondary violarite and cubanite (Chai and Naldrett 1992a; Yang et al. 2006). Song et al. (2009) suggested that the ore bodies at Jinchuan were produced by injection of sulfide-rich and sulfide-poor olivine mushes originating in a deep staging chamber. Following contamination in the lower crust, the magma is thought to have been intruded as a sill crosscutting gneisses and marbles (Lehmann et al. 2007). They report common inclusions of marble, now decarbonated and converted to diopside-rich skarn, near the contact and dispersed through much of the intrusion suggesting interaction of the hot magma with the host floor marbles. Parts of the intrusion were invaded by CO_2-rich fluids which percolated upwards into the intrusion. Post-emplacement, the intrusion and its host rocks underwent greenschist facies alteration (Lehmann et al. 2007). This metamorphism and local shear deformation is evidenced by the presence of schist fabrics and an abundance of typical greenschist facies hydrothermal minerals (Chai and Naldrett 1992b; Li et al. 2004; Naldrett 2004; Zhou et al. 2002; Yang et al. 2006). The intrusion and ores have been subjected to postmagmatic processes to varying degrees resulting in a mix of fresh magmatic, hydrothermally altered, and sheared ores.

The age of the Jinchuan intrusion is somewhat controversial. It was previously believed that the intrusion is

1,500 Ma from K-Ar dating and crosscutting relationships however, new U-Pb dating appears to show that the intrusion is in fact ~825 Ma (Li et al. 2005). This age has been further constrained by Zhang et al. (2010) who used U-Pb dating on zircon and baddeleyite from a lherzolite sample to obtain an age of 831.8±0.6 Ma.

Analytical methods

Mineralogical studies were performed on one thin section from each of 16 samples and a further five thin sections of the freshest and most PGE-enriched sample JZ-04. Petrography was examined using conventional optical microscopy and minerals were analyzed using a Cambridge Instruments (now Carl Zeiss NTS) S360 scanning electron microscope (SEM; Cardiff University, UK). Polished sections were searched for PGM systematically using the SEM set at a magnification of ×100. Quantitative analyses of the larger PGM were obtained using an Oxford Instruments INCA Energy EDX analyzer attached to the SEM. Operating conditions for the quantitative analyses were 20 kV, with a specimen calibration current of ~1 nA and a working distance of 25 mm. A cobalt reference standard was regularly analyzed in order to check for any drift in the analytical conditions. A comprehensive set of standards obtained from MicroAnalysis Consultants Ltd. (St Ives, Cambridgeshire, UK) was used to calibrate the EDX analyzer. Images were obtained using a four-quadrant backscattered detector operating at 20 kV, a beam current of ~500 pA, and a working distance of 13 mm under which conditions, magnifications of up to ×15,000 are possible. To place the PGM in context mineralogically, the associated silicate and sulfide minerals were identified qualitatively using the SEM.

Four samples displaying different degrees of alteration were selected for LA-ICP-MS. Analyses were carried out using a New Wave Research UP213 UV laser system coupled to a Thermo X Series ICP-MS (Cardiff University, UK). Details of the procedures for the laser analysis follow those given in Holwell and McDonald (2007). PGE and other elements were determined in time-resolved analyses mode (time slices of 350 ms) as the laser beam followed a line designed to sample different sulfide phases. The beam diameter employed was 30 μm with a frequency of 10 Hz; the sample was translated at 6 μm/s relative to the laser. Acquisitions lasted between 80 and 400 s, and a gas blank was measured for 30–40 s prior to the start of the analysis. Sulfur concentrations were measured prior to LA-ICP-MS using the SEM and ^{33}S was used as an internal standard. Subtraction of gas blanks and internal standard corrections were performed using Thermo Plasmalab software.

Calibration was performed using a series of synthetic Ni-Fe-S standards prepared from quenched sulfides. The standards incorporate S, Ni, Fe, and Cu as major elements and Co, Zn, As, Se, Ru, Rh, Pd, Ag, Cd, Sb, Te, Re, Os, Ir, Pt, Au, and Bi as trace elements. The compositions of the five sulfide standards are given in Table 1. Major elements were determined by SEM and trace elements by solution ICP-MS after dissolution of between two and five individual sulfide spherules from each class of standard. Sulfide spherules were mounted in resin and polished to produce the laser standards. Sulfide homogeneity was checked using the SEM; this revealed complete homogeneity in Fe-Ni-S Standards 1, 2, and 5 and only minor separation of blebs of Cu-rich sulfide (1–2 μm across) in Standards 2 and 3 that contain Cu as a major element. The standards produce five-point calibration curves for S, Ni, and Fe. Standards 1, 4, and 5 produce three-point calibration curves for PGE, Ag, Cd, Re, Au, and semimetals. Standards 1, 2, and 3 produce three-point calibration curves for Cu, Co, and Zn and matrix-matched corrections for argide species ($^{59}Co^{40}Ar$, $^{61}Ni^{40}Ar$, $^{63}Cu^{40}Ar$, $^{65}Cu^{40}Ar$, and $^{66}Zn^{40}Ar$) that interfere with light PGE isotopes. Corrections for ^{106}Cd on ^{106}Pd and ^{108}Cd on ^{108}Pd were determined using Standard 1 but Cd concentrations in the Jinchuan sulfides were so low as to make these trivial in most unknown samples. Argide- and isobaric-corrected data are indicated by asterisks beside ^{99}Ru, ^{103}Rh, ^{106}Pd, and ^{108}Pd in Table 6. Where independent corrections have been applied to different isotopes of the same element (e.g.,

Table 1 Compositions of quenched sulfide standards used for LA-ICP-MS

		Std-1	Std-2	Std-3	Std-4	Std-5
S	wt%	29.50	30.1	28.60	31.60	30.1
Fe	wt%	5.18	7.3	4.15	9.47	10.6
Ni	wt%	62.37	51.7	46.90	59.03	58.3
Cu	wt%	–	11	19.81	–	–
Co	ppm	36	4,850	15,000	–	–
Zn	ppm	100	3,000	4,600	–	–
As	ppm	–	–	–	57	108
Se	ppm	–	–	–	140	264
Ru	ppm	–	–	–	51	160
Rh	ppm	–	–	–	51	160
Pd	ppm	–	–	–	50	120
Ag	ppm	–	–	–	147	152
Cd	ppm	143	–	–	–	–
Sb	ppm	–	–	–	52	108
Te	ppm	–	–	–	210	644
Re	ppm	–	–	–	61	127
Os	ppm	–	–	–	50	160
Ir	ppm	–	–	–	55	160
Pt	ppm	–	–	–	50	125
Au	ppm	–	–	–	44	125
Bi	ppm	6	6.5	5	146	263

Dashes indicate that the element was not added to the standard

$^{66}Zn^{40}Ar$ on ^{106}Pd and ^{108}Cd on ^{108}Pd) the independently corrected values vary by less than 20 % (and commonly <5 %) indicating that the corrections are robust. The accuracy of the LA-ICP-MS procedure for PGE was checked by analysis of the Laflamme-Po724 standard run as an unknown against the Cardiff sulfide standards and results of this given along with the unknown samples in Table 6.

Traditionally, PGM are located in polished thin sections but recently a method using a hydroseparator has been developed which involves extracting PGM from crushed samples (e.g., Cabri et al. 2008, 2009). However, this method loses most of the textural relationships between the PGM and their host minerals. The study presented here required that the relationships with the hosts of the PGM were preserved so that a paragenetic sequence of PGE-bearing mineral formation could be identified. The numbers of PGM located and the consistency of PGM hosts and relationships in the altered samples suggests that the PGM located are representative of the PGM present in the ore samples studied.

Whole rock Ir, Ru, Rh, Pt, and Pd analyses are taken from Song et al. (2006; Table 2) and analytical details are given there. The same powders were analyzed at Genalysis in Perth Western Australia as a check and at Genalysis PGE concentrations were determined using a Ni collection fire assay technique followed by ICP-MS analysis. Gold and Os were also determined in this second set of analyses.

Results

PGE concentrations and petrology

The samples studied are a subset of those described in Song et al. (2006) in which PGE and trace element geochemistry

Table 2 Whole rock PGE analyses in ppb

Sample	Ore body	Au	Os	Ir	Ru	Rh	Pt	Pd	Pt/Pd	No of PGM	% fresh	% BMS	Rock type
JZ-04	#1	170	94	92	73	42	3,059	164	18.65	2	70	25	Dun
				56	67	41	3,972	165	24.07				
JC-29	#1	105	4	7	7	4	55	60	0.92	5	35	5	Lhe
				3	5	6	70	57	1.23				
JZ-02	#1	–	–	37	26	22	6.9	128	0.05	2	25	30	Dun
JZ-05	#1	92	21	24	21	13	191	128	1.49	29	25	5	Lhe[a]
				8	5	8	19	46	0.41				
JC-24	#2	133	14	19	15	10	95	80	1.19	17	35	3	Lhe
				16	14	8	74	71	1.04				
JC-54	#2	43	3	4	4	3	118	16	7.38	0	30	7	Lhe
				3	3	2	79	15	5.27				
JC-03	#24	66	13	19	13	8	178	93	1.91	5	70	1	Lhe[a]
				14	14	7	196	97	2.02				
JC-05	#24	–	–	4	5	2	58	33	1.76	29	50	1	Lhe[a]
JC-30	#24	81	26	30	20	19	423	261	1.62	21	30	10	Dun
				8	20	15	336	268	1.25				
JC-23	#24	149	25	33	24	13	285	135	2.11	2	10	5	Lhe
				25	25	12	282	134	2.10				
JC-26	#24	–	–	13	19	11	114	138	0.83	6	10	20	Lhe
JC-20	#24	64	14	15	12	8	120	77	1.56	14	0	30	Lhe[a]
				10	11	6	74	73	1.01				
JC-38	#24	–	–	17	20	16	37	219	0.17	23	0	30	Dun
JC-39	#24	–	–	110	122	44	33	318	0.10	3	0	20	Dun
JC-40	#24	–	–	12	31	15	9	168	0.05	3	0	40	Dun
JC-49	#24	–	–	89	260	237	55	532	0.10	22	0	100	Mas[a]

PGE data from Prichard et al. (2005) was analyzed at Genalysis Plc., Perth, Australia and detection limits for the Genalysis results are 2 ppb as are those analyses with all 6 PGE and Au. PGE data minus Os and Au are from Song et al. (2006). Samples are listed by ore body and then increasing in alteration downwards. Rock types are identified from thin section petrography. The “% fresh” is based on the alteration of the silicates

– not analyzed, *Dun* dunite, *Lhe* lherzolite, *Mas* massive BMS

[a] Classified in Song et al. (2006)

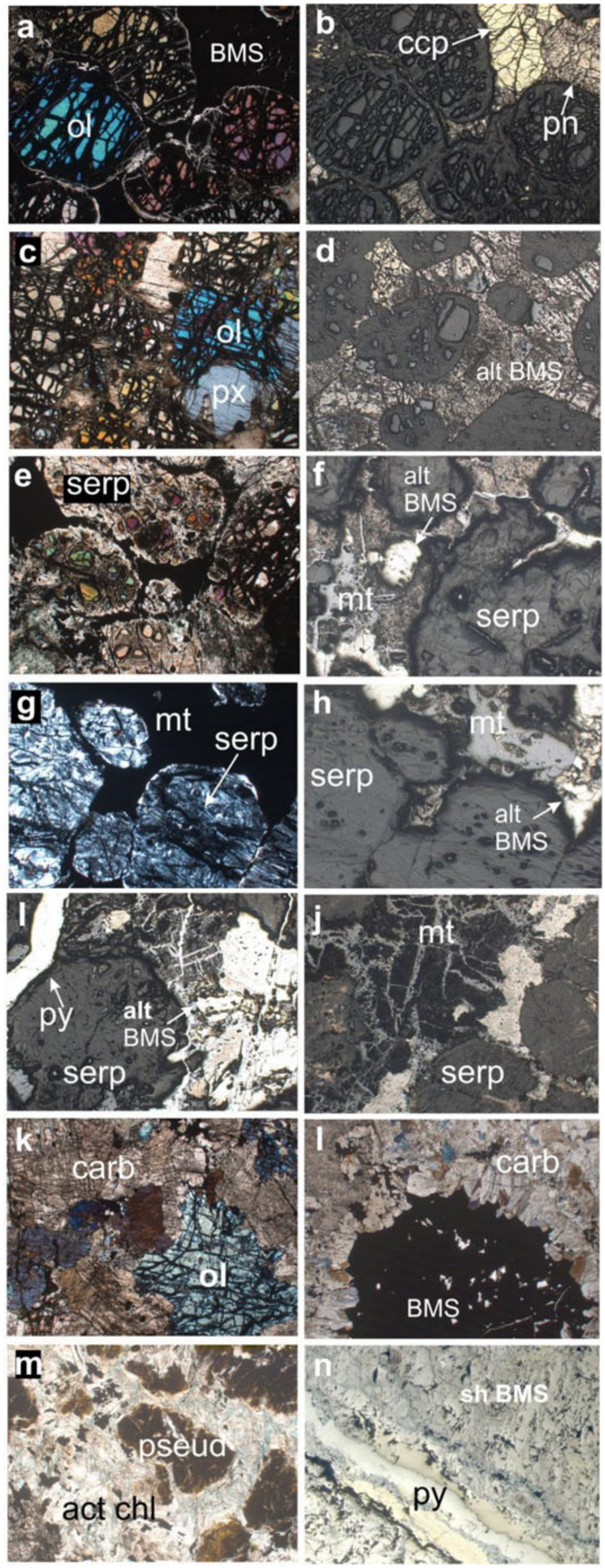

Fig. 2 Photomicrographs of samples showing silicate and BMS petrography arranged in increasing alteration from **a** to **n**. Width of photomicrograph in all cases represents 3.5 mm. **a** View in cross-polarized light (*xpl*) of JZ-04 showing 70 % fresh olivine (*ol*), **b** same view in reflected light showing interstitial fresh BMS chalcopyrite (*ccp*) and pentlandite (*pn*), **c** 70 % fresh olivine and orthopyroxene (*px*) in JC-03, **d** reflected light view of JZ-02 showing 25 % fresh olivine and altered (*alt*) BMS, **e** view of JZ-05 in xpl showing serpentinised (*serp*) olivine with variable amounts of relict olivine in different grains with black interstitial BMS, **f** reflected light view of JC-30 showing serpentinised olivine, altered interstitial BMS and magnetite (*mt*) veining, **g** view in xpl of JC-38 showing 100 % serpentisation of olivine with interstitial BMS, **h** same view in reflected light showing altered BMS and magnetite, **i** reflected light view of JC-39 showing serpentinised olivine with interstitial altered BMS with magnetite veining and late veins of pyrite (*py*), **j** reflected light view of JC-40 showing extensive alteration of BMS with sulfur loss resulting in magnetite veining, **k** view in xpl of JC-29 showing fresh olivine surrounded by carbonate (*carb*), **l** view in xpl of JC-39 showing extensive alteration to carbonate surrounding BMS, **m** plane polarized light view of JC-20 showing pseudomorphed (*pseud*) olivine surrounded by fibrous actinolite and chlorite (*act chl*), **n** reflected light view of JC-49 showing sheared (*sh*) texture of massive BMS with late cross cutting veins of pyrite

Table 3 Distribution and abundance of PGM listed by sample and arranged in increasing percentages of alteration downwards. Each type of PGM is characterized by number (no), area of exposure on the 2D thin section in square micrometer (area) and average area in micrometer (ave)

	$Pd\text{-}Bi_2$			Pd or Pt-Bi-Te			Pd-Bi-Se ± Te			$PtAs_2$			Ir-Rh-Pt-As-S			Others		Fresh	BMS
	No	Area	Ave	No	Area	Ave	No	Area	Ave	No	Area	Ave	No	Area	Ave	No	Area	%	%
JZ-04													2	64	32.0			70	25
JC-03				2	15	7										2	10.5	70	1
JC-05	12	31.5	2.6	2	19.0	9.0	1	0.3	0.3	8	63	7.8	3	23	7.7	4	35.3	50	1
JC-29	3	5.0	1.7	2	3.0	1.5												40	5
JC-24	2	3.0	1.5	13	123.0	8.1										2	17.0	35	3
JZ-05																		30	5
JC-54																		30	7
JC-30	11	76.5	7.0	2	21.0	10.5	6	35.5	5.9				1	10	10.0			30	10
JZ-02													2	20	10.0			30	30
JC-23				1	2.0	2.0				1	15	15.0						10	5
JC-26				2	54.0	27.0	1	15.0	15.0							2	3.0	10	20
JC-39	3	8.0	2.7										3	52	17.0			0	20
JC-38	6	62.0	10.3	3	26.0	8.7	14	62.5	4.5									0	30
JC-20	7	23.5	3.4	3	2.5	0.8	2	16.0	8.0				1	8	8.0	1	6.0	0	30
JC-40													3	37	12.0			0	40
JC-49													18	443	25.0			0	100
Total	44	209.5	4.8	30	265.5	8.1	24	129.3	5.4	9	78	8.6	33	657	20.7	11	71.8		

Others include a 3×3 μm Pt(PdAg)S(Cu) and a 3×1 μm Pd-Te in JC-03, a 1×0.5 μm Pt-Te and 3 Au-Pd ave. 3×3 μm in JC-05, a Pd-Pb-Bi-Te-Se 3×2 μm in JC-20, 2 Au-Ag-Pd-Bi-Te ave. 3×2 μm in JC-24 and two Pd-Bi-Sb ave. 2×1 μm in JC-26. JC-24 contains two electrum ave. 2×1 μm and JC-29 contains one electrum 1×0.5 μm

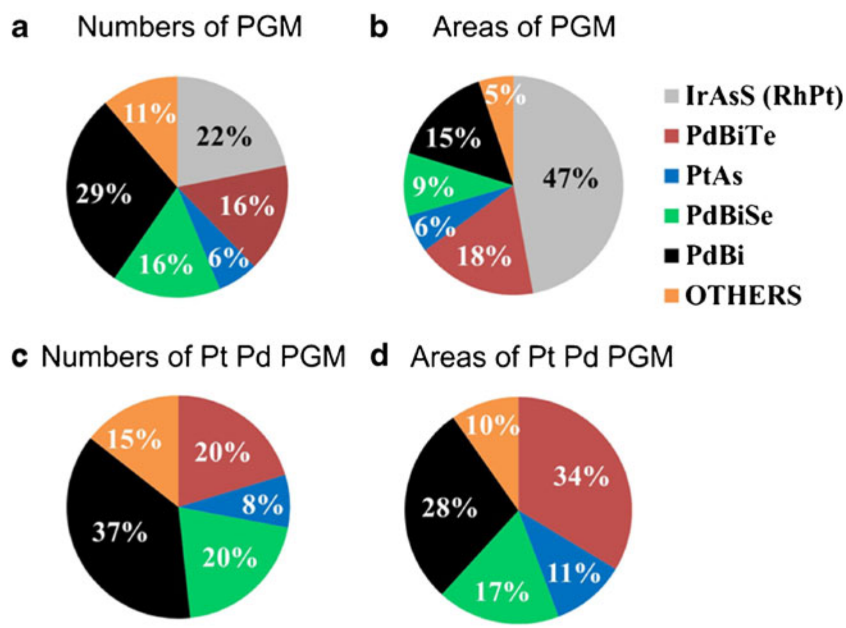

Fig. 3 **a** Numbers of different types of PGM, **b** areas (in square micrometer) of different types of PGM, **c** numbers of different types of Pt-Pd-bearing PGM, **d** areas (in square micrometer) of different types of Pt-Pd-bearing PGM. Others include PtBiTe and Au grains

is described and discussed in detail. The samples have a wide range of PGE concentrations with JZ-04 containing the highest concentrations of Pt (3,059 ppb) but also the highest Os (94 ppb), Ir (92 ppb), Ru (73 ppb), and Rh (42 ppb; Table 2). Both the fresher and more altered samples have general positive slopes on chondrite normalized trends (Song et al. 2006). The results presented here build on the study by Song et al. (2006) by using these already characterized samples to make a detailed study of the distribution of the PGE between BMS and PGM. An understanding of this PGE distribution relies on placing them in context with their silicate, BMS, and oxide mineral hosts. The samples consist of a suite of lithologies that show a range of alteration of silicates from 70 % fresh to 100 % altered with varying amounts of

Table 4 Selective representative quantitative analyses of PGM

	1 JC-05	2 JC-26	3 JC-30	4 JC-38	5 JZ-02	6 JZ-04
Fe	3.53	–	3.08	–	–	–
Ni	–	5.82	–	–	–	–
Pd	21.07	15.19	21.27	21.68	–	–
Rh	–	–	–	–	1.01	–
Ir	–	–	–	–	52.34	54.26
Pt	–	–	–	–	8.13	9.64
S	–	–	–	–	10.67	10.69
As	–	–	–	–	25.94	26.36
Te	–	30.18	–	–	–	–
Bi	75.97	49.06	76.91	78.61	–	–
Totals	100.57	100.24	101.27	100.29	98.09	100.95
Analysis	Sample No.	PGM		Composition		
1	JC-05 A2	Froodite		$Pd_{0.95}(Bi_{1.75}Fe_{0.30})_{2.05}$		
2	JC-26 A1	Ni-Michenerite		$(Pd_{0.60}Ni_{0.42})_{1.02}\ Bi_{0.99}Te_{1.00}$		
3	JC-30 B4	Froodite		$Pd_{0.96}(Bi_{1.77}Fe_{0.27})_{2.04}$		
4	JC-38 A2	Froodite		$Pd_{1.05}Bi_{1.95}$		
5	JZ-02 A1	Pt-Rh-Irarsite		$(Ir_{0.81}Pt_{0.12}Rh_{0.03})_{0.96}As_{1.04}S_{1.00}$		
6	JZ-04 D1	Pt-Irarsite		$(Ir_{0.83}Pt_{0.15})_{0.98}As_{1.04}S_{0.98}$		

Analyses given in weight percent, sample numbers in which PGM occur are given, e.g., JC-05a

– not detected

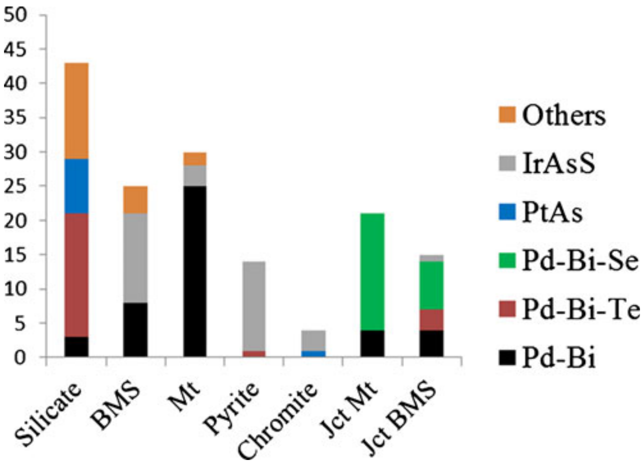

Fig. 4 Histogram showing the host associations of the different PGM and Au-bearing grains Jct = junction, Mt = magnetite. Others include Au-bearing grains and Pt-Bi-Te

BMS and PGE (Table 2). Samples are classified into dunite and lherzolite based on geochemistry given in Song et al. (2006) and the petrology of the thin sections. The samples were collected from ore bodies 1, 2, and 24 (Table 2) which account for more than 90 % of the Ni, Cu, and PGE in the Jinchuan deposit (Chai and Naldrett 1992b).

Except for JC-49, which consists of sheared massive BMS, all the other samples are undeformed and display varying degrees of alteration. The textures and different stages of alteration of the samples are illustrated in Fig. 2a–n. Cumulate textures are dominant with fresher samples of dunite-containing olivine or in the case of lherzolite, olivine, and pyroxene with altered interstitial plagioclase (Fig. 2a–c). Minor chrome spinel is present in some samples, but is usually extensively altered to magnetite. Most sections show variable alteration across them with relict olivine and pyroxene in some areas and not others; this variation in alteration may be considerable even over short distances (e.g., Fig. 2e). The igneous textures are clearer in fresher samples and olivine is the least altered silicate mineral (e.g., in JZ-04 and JC-03; Fig. 2a–c). In some cases, igneous textures are perfectly preserved with serpentine-filled olivine pseudomorphs (e.g., JC-38; Fig. 2g). In samples that are completely altered, the original mineralogy and textures are in some cases difficult to distinguish masked by the development of abundant actinolite and chlorite (e.g., JC-20; Fig. 2m). In some samples, the alteration includes the presence of abundant carbonate. For example, this is the case for JC-29 and JC-39 which still preserve very fresh olivine (Fig. 2k and l). These samples contain elevated Sr with 97 ppm in JC-29 compared with 9 ppm in JZ-02 which is free of carbonate in thin section (Sr values from Song et al. (2006)). This compares with values in unaltered lherzolites of less than 4 ppm (Gruau et al. 1998).

The samples have variable amounts (1–40 %; Table 2) of interstitial BMS (pentlandite, pyrrhotite, and chalcopyrite) resulting in a typical and highly distinctive net-textured ore that characterizes the Jinchuan deposit. In JZ-04 (the freshest sample with over 3 ppm Pt; Fig. 2a and b), the BMS are fresh and homogeneous taking a smooth polish. In all the other sections, the BMS are altered with, for example, pentlandite being partially oxidized and having a mottled appearance (e.g., Fig. 2d). In the more altered samples, the oxidized BMS are frequently rimmed (e.g., Fig. 2h) and veined by secondary magnetite (e.g., JC-30, JC-39, and

Table 5 Numbers of different PGM and Au grains in different host minerals

		Au grains	IrAsS	$PdBi_2$	PdBiTe	PtBiTe	PtTe	$PtAs_2$	PdBiSb	PdBiSe
Enclosed	cpy		3	1		1				
Enclosed	po	1	5	7	2					
Enclosed	pn		5			2				
Enclosed	Py		13		1					
Enclosed	chr-spinel	3					1			
Enclosed	mt		3	25		2				
Enclosed	silicate	8		3	16	2	2	8		
Junction	silicate/BMS				2				2	
Junction	mt/BMS			4						17
Junction	cpy/pn		1							3
Junction	cpy/po				1					1
Junction	cpy/Zn			1						
Junction	po/pn			3	2					
Junction	pn/pn									2
Junction	cpy/cpy									1
Total		9	33	44	24	7	2	9	2	24

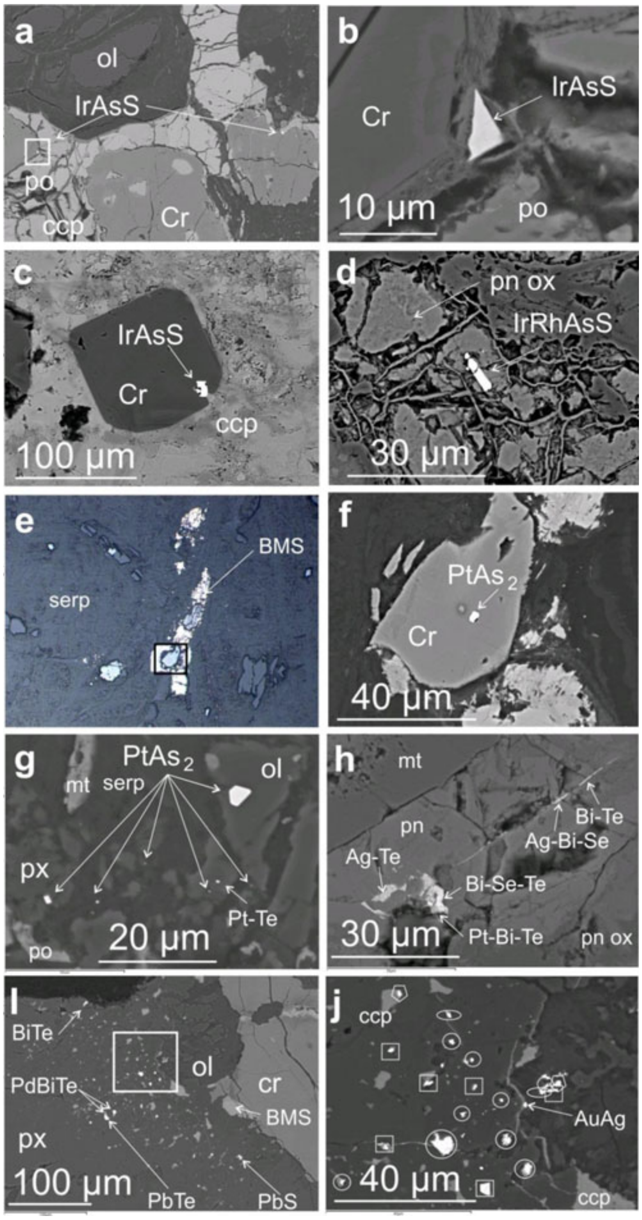

Fig. 5 Back-scattered electron (BSE) images of PGM. **a** BSE image showing the location of two irarsite (IrAsS) crystals at the junction of chromite (*Cr*) and BMS (pyrrhotite po and chalcopyrite ccp) interstitial to chromite and partially serpentinised olivine (*ol*) in JZ-02. **b** Close up of the *box* marked on **a** showing one of the irarsite grains. **c** BSE image of an irarsite grain enclosed in chromite surrounded by sheared BMS including chalcopyrite in JC-49. **d** BSE image of an elongate cracked and broken IrRhAsS surrounded by altered and veined pentlandite (*pn ox*) also in JC-49. **e** Reflected light photomicrograph of a chromite grain surrounded by interstitial altered BMS between serpentinised olivine (*serp*) in JC-23. **f** BSE image of a close up of the box in **e** showing a sperrylite ($PtAs_2$) enclosed in chromite. **g** BSE image of a cluster of sperrylite grains in partially serpentinised olivine and partially altered pyroxene (*px*) with magnetite (*mt*) in JC-05. **h** BSE image of a cluster of tellurides and selenides in altered pentlandite and along a vein traversing the pentlandite in JC-20. **i** BSE image of a cluster of PGM at the altered junction between olivine and pyroxene in JC-24. **j** Close up of the *box* in **i** showing gold and PGE-bearing minerals with *squares* around Au, Ag, Pd, Bi, tellurides; *pentagons* around Pb-Te; *circles* around Pd-Bi-Te; and *elipses* around $PdBi_2$

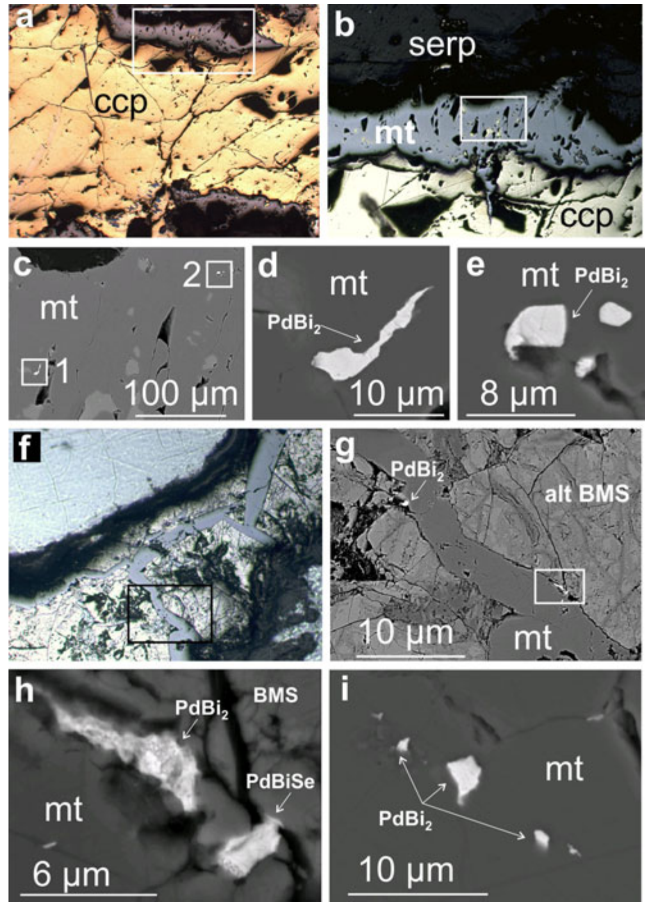

Fig. 6 **a** Reflected light photomicrograph of chalcopyrite (*ccp*) showing sulfur loss and formation of magnetite at the edges in JC-38. **b** Close up of box in **a** showing the magnetite (*mt*) selvage at the edge of the chalcopyrite and adjacent to serpentine (*serp*). **c** Close up of box in **b** showing a BSE image of magnetite enclosing froodite ($PdBi_2$). **d** Close up of box 1 in **c** showing $PdBi_2$ and **e** close up of box 2 in **c** showing a cluster of PdBi. **f** Reflected light photomicrograph of magnetite veins crossing altered BMS interstitial to altered olivine. **g** Close up of box in **f** showing a BSE photomicrograph of the magnetite vein cross-cutting altered BMS showing the position of two $PdBi_2$ minerals at the edge of the magnetite vein. **h** Close up of box in **g** showing one of the $PdBi_2$ at the junction between the magnetite and BMS partially altered to Pd-Bi selenide. **i** Row of $PdBi_2$ enclosed in magnetite in JC-39

JC-40; Fig. 2f and j). Late pyrite-filled veins are also a feature of the more severely altered samples (e.g., JC-39 and JC-49; Fig. 2i and n).

PGM types and sizes

Despite the freshest sample, JZ-04 having the highest concentration of PGE with over 3 ppm Pt, the PGM host for the Pt has not been found. Initial searches of a polished block revealed two irarsite grains (Table 3). A further extensive search of five more polished sections cut from the same slice of sample revealed five more irarsite grains containing

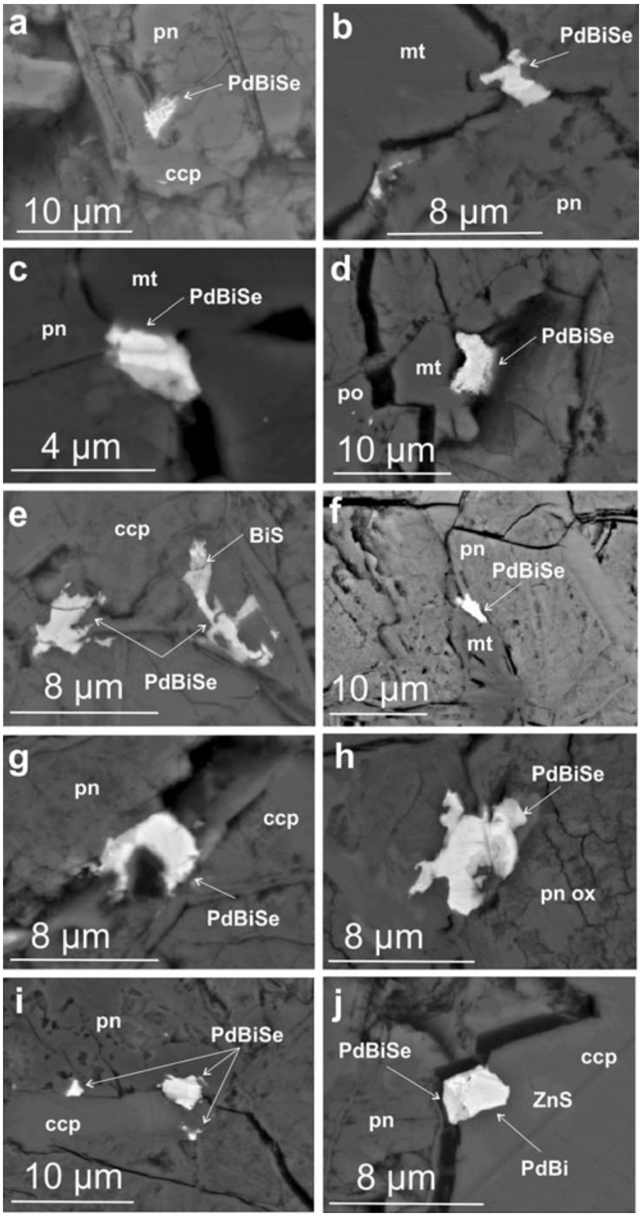

Fig. 7 Back scattered electron images of padmaite (PdBiSe) at junctions between magnetite and BMS and along cracks through BMS, **a** from JC-26, **b**–**d** from JC-30, **e**–**j** from JC-38

consistently between 11 and 13 % Pt and one PdBi selenide. The irarsite is present in pentlandite, chalcopyrite, silicates, and in two cases is surrounded by gersdorffite. JC-03 is also 70 % fresh and also only contains two PGM but is not enriched in PGE with a Pt + Pd content of only 271 ppb. In contrast, over 140 PGM grains have been located in the more altered samples and their abundances in terms of numbers and areas as exposed in 2D on the thin section surface are listed in Table 3.

The PGM identified are predominantly Pd-bearing with 44 froodite ($PdBi_2$), 24 michenerite ([Pd,Pt]BiTe), and 24 padmaite (PdBiSe), an unusual Se-bearing PGM. Others include 33 PGM rich in Ir and Rh belonging to the irarsite–hollingworthite–platarsite ([Ir-Rh-Pt]AsS) solid solution series and nine $PtAs_2$. The remaining PGM identified are much less common and make up only 11 % of the total PGM observed (Fig. 3). These include Pd-Te, Pt-Te, and Pd-Bi-Sb. A few Au-bearing minerals include three grains of Au-Pd and three grains of Au-Ag in JC-29. The froodite has an average diameter of 4.8 μm with the largest two grains being 15 and 8 μm in diameter. All the remaining 42 grains are 5 μm or less. The michenerite grains observed are slightly larger with an average diameter of 8.1 μm; the largest two grains are 14 and 8 μm in diameter. Padmaite is smaller, averaging 5.4 μm with the largest being 15 μm in diameter. The irarsite averages 5 μm in diameter, whereas the largest is 12 μm and sperrylite averages 2.1 μm in diameter with the largest grain being 7 μm. Thus, the PGM are all small and the vast majority are less than 5 μm in diameter. To give a better indication of the relative volumes of the different PGM, the area of each PGM was calculated (Table 3 and Fig. 3).

The total number of each type of PGM observed compared with areas of these PGM highlights the impact of PGM size and shows which phases are potentially more important in collecting PGE. For example, irarsite, being slightly larger, has a greater percentage of the total PGM calculated by area (47 %) than as a percentage of the total number of PGM (22 %; Fig. 3a and b). When just the Pt- and Pd-bearing PGM are considered, the proportions of the PGM by number and by area are similar, although froodite has a greater percentage of the total number of PGM (37 %) whereas michenerite has a greater percentage of the total PGM in terms of area (34 %; Fig. 3c and d). Selective representative quantitative analyses of PGM are presented in Table 4.

PGM host associations

The host associations for the different PGM are very distinctive (Fig. 4 and Table 5). The main hosts are silicates, magnetite, pyrite, chrome-spinel, and BMS (taken as one category including pentlandite, pyrrhotite, and chalcopyrite). Another textural position for the PGM is at mineral boundaries principally between magnetite and BMS or between different individual BMS.

Irarsite is different from the other PGM as it is predominantly enclosed in BMS or pyrite. It is the only PGM located in the very PGE-enriched JZ-04. It also occurs in JZ-02 at the contact of chrome spinel and BMS (Fig. 5a and b) and is identified enclosed by chromite (Fig. 5c). It is the dominant PGM in pyrite in the massive sheared ore (JC-49) where it may be broken and cracked (Fig. 5d) and is enclosed in oxidized altered pentlandite.

Sperrylite, Pd-Bi telluride, as well as rare Au-bearing PGM and electrum are almost entirely enclosed by or

Table 6 Laser ablation-ICP-MS data from BMS in fresher samples JZ04, more altered sample JZ-02 and very altered samples JC-26 and JC-38

Sample	Mineral	S wt%	57Fe %	59Co ppm	61Ni %	65Cu %	66Zn ppm	75As ppm	82Se ppm	99Ru[a] ppm	103Rh[a] ppm	106Pd[a] ppm	108Pd[a] ppm	109Ag ppm	111Cd ppm	121Sb ppm	125Te ppm	185Re ppm	189Os ppm	193Ir ppm	195Pt ppm	197Au ppm	209Bi ppm	S/Se
JZ-04	Cpy	34.9	34.17	149	0.77	27.31	11620	<1.2	126	0.13	<0.1	<0.15	<0.2	2.59	22.71	0.62	6.52	0.15	0.93	0.01	<0.01	<0.01	0.7	2770
Altered		34.9	35.16	224	1.14	22.41	12860	<1.2	176	<0.05	<0.1	<0.15	<0.2	10.40	19.03	2.16	9.92	0.02	0.04	<0.01	0.10	0.13	0.8	1983
30 %		34.9	32.33	73	0.38	25.21	18310	<1.2	105	<0.05	<0.1	0.66	0.73	9.55	26.08	0.48	5.35	<0.02	<0.01	<0.01	0.00	<0.01	4.9	3324
JZ-02		34.5	31.04	403	1.93	27.84	4711	<1.2	89	0.14	<0.1	0.16	0.14	4.82	3.47	0.92	3.64	0.82	<0.01	<0.01	<0.01	0.03	1.3	3876
Altered		34.5	31.85	<4	0.02	33.50	46970	<1.2	86	<0.05	<0.1	<0.15	<0.2	1.84	27.85	1.03	7.11	0.06	0.07	0.05	<0.01	0.08	1.4	4012
75 %		34.5	27.41	235	1.19	22.93	2525	1.5	203	<0.05	<0.1	<0.15	<0.2	5.01	2.55	2.36	6.50	0.10	<0.01	0.01	0.04	0.06	1.3	1700
JC-26		34.5	31.27	373	2.29	26.74	3195	1.7	129	<0.05	<0.1	0.62	0.64	22.70	2.86	<0.25	15.00	0.02	0.06	0.02	<0.01	<0.01	7.1	2674
Altered		34.5	27.79	330	1.91	24.78	23130	<1.2	129	0.09	<0.1	<0.15	<0.2	20.00	23.27	0.56	54.60	<0.02	<0.01	<0.01	<0.01	<0.01	2.8	2674
90 %		34.5	28.61	<4	<0.08	23.37	1419	1.6	102	0.36	<0.1	0.46	0.46	19.20	4.70	<0.25	3.63	<0.02	<0.01	<0.01	<0.01	<0.01	2.3	3382
JC-38		35	29.84	<4	<0.08	27.92	2079	<1.2	155	<0.05	<0.1	<0.15	<0.2	7.62	3.59	<0.25	12.90	0.11	0.03	<0.01	0.06	<0.01	0.9	2258
Altered		35	28.34	<4	<0.08	26.71	3156	<1.2	149	0.19	<0.1	<0.15	0.25	5.73	4.17	0.64	9.43	0.05	0.03	<0.01	0.06	0.08	2.0	2349
100 %		35	39.79	<4	<0.08	21.81	143	<1.2	146	<0.05	<0.1	<0.15	<0.2	2.07	<0.90	0.37	1.52	0.68	0.18	0.10	0.06	<0.01	0.2	2397
		35	36.90	372	1.57	16.12	2189	<1.2	137	<0.05	<0.1	0.16	<0.2	3.15	1.72	0.79	<0.90	<0.02	<0.01	<0.01	<0.01	0.11	0.2	2555
JZ-04	Pn	33	50.54	6390	28.82	0.77	257	3.1	109	0.30	<0.1	1.96	2.01	10.80	<0.90	0.58	7.81	0.17	0.42	0.01	<0.01	<0.01	1.8	3028
Altered		33	42.16	6134	27.82	1.44	843	<1.2	160	0.57	<0.1	1.90	2.03	5.28	1.49	1.58	4.57	0.05	0.57	0.01	0.02	0.05	1.2	2063
30 %		33	38.94	6271	29.45	0.06	67	<1.2	91	0.53	<0.1	1.30	1.86	4.88	<0.90	0.92	2.23	0.13	0.18	0.02	0.02	<0.01	1.2	3626
		33	40.53	6701	31.10	0.37	261	3.6	102	0.50	<0.1	2.63	2.58	9.56	0.32	0.51	13.70	0.03	<0.01	<0.01	<0.01	<0.01	0.4	3235
JZ-02		33	35.33	5660	29.06	1.26	541	1.7	179	0.45	<0.1	0.99	0.72	11.60	<0.90	1.62	8.54	0.43	0.25	0.02	<0.01	0.12	0.9	1844
Altered		33	44.13	6354	32.23	0.42	853	14.2	123	0.34	0.27	1.53	1.15	14.30	<0.90	0.72	13.75	0.08	0.39	0.02	0.03	0.05	3.6	2683
75 %		33	41.23	6524	31.35	0.06	320	<1.2	174	<0.05	<0.1	1.34	1.40	5.64	<0.90	1.42	2.52	0.24	0.32	0.04	<0.01	0.07	1.3	1897
		33	46.71	6740	31.61	1.04	2457	2.3	104	0.46	0.17	1.46	1.60	13.20	<0.90	0.84	7.48	0.46	0.23	0.01	<0.01	0.02	4.0	3173
		33	40.39	5873	29.42	0.05	308	<1.2	159	0.56	<0.1	1.67	1.67	3.40	<0.90	1.61	3.73	0.48	0.29	0.02	0.02	0.03	0.9	2075
JC-26		33	25.38	5507	29.05	0.46	182	9.8	201	0.14	<0.1	0.72	0.62	62.70	<0.90	1.43	33.50	<0.02	<0.01	<0.01	<0.01	0.01	2.1	1642
Altered		33	24.04	5442	27.39	0.63	242	2.6	199	0.18	<0.1	0.64	0.55	58.50	<0.90	1.38	32.30	<0.02	<0.01	<0.01	<0.01	0.04	5.0	1658
90 %		33	23.32	5921	26.05	0.11	94	8.2	172	0.08	<0.1	3.82	3.67	19.80	<0.90	1.30	3.49	0.12	0.02	<0.01	<0.01	0.04	0.9	1919
		33	28.89	5694	27.14	0.04	117	8.1	194	0.29	<0.1	2.91	3.03	16.50	<0.90	1.63	12.40	0.50	0.34	0.01	<0.01	0.03	0.9	1701
		33	27.90	5786	26.55	<0.05	70	40.9	192	0.66	<0.1	4.20	4.25	57.10	<0.90	2.04	7.92	1.06	0.36	0.01	<0.01	0.59	12.2	1719
JZ-04	Po	39.6	56.75	288	1.34	<0.05	85	<1.2	164	0.37	<0.1	<0.15	<0.20	1.02	<0.90	1.31	1.36	0.30	0.58	<0.01	<0.01	0.03	0.9	2415
Altered		39.6	62.62	15	0.10	0.11	60	<1.2	105	<0.05	<0.1	<0.15	<0.20	2.64	<0.90	0.73	<0.90	<0.02	<0.01	<0.01	0.02	<0.01	2.3	3771
30 %		39.6	63.18	47	0.24	2.01	489	<1.2	150	<0.05	<0.1	<0.15	0.22	3.62	1.24	1.52	3.00	<0.02	<0.01	<0.01	0.04	0.03	2.9	2640
		39.2	56.46	84	0.99	<0.05	<40	<1.2	149	0.29	<0.1	<0.15	<0.20	1.74	<0.90	0.76	<0.90	0.11	0.27	<0.01	<0.01	<0.01	0.4	2631
JZ-02		38.5	60.34	9	0.12	0.76	533	<1.2	149	0.25	<0.1	<0.15	<0.20	0.76	<0.90	1.39	1.26	0.13	0.29	0.00	0.02	0.02	0.8	2584
Altered		38.5	61.12	331	1.75	0.21	462	<1.2	139	0.35	<0.1	<0.15	<0.20	1.26	<0.90	1.34	0.97	0.54	0.38	<0.01	<0.01	0.03	1.0	2770
75 %		38.5	62.80	40	0.37	<0.05	93	<1.2	120	0.20	<0.1	<0.15	<0.20	0.91	<0.90	0.70	<0.90	0.42	0.24	0.03	<0.01	<0.01	1.2	3208
		38.5	59.50	185	0.87	<0.05	117	<1.2	123	0.28	<0.1	<0.15	0.26	0.58	<0.90	0.68	<0.90	0.43	0.25	0.03	0.02	<0.01	1.5	3130
		39.6	62.74	229	1.33	0.13	110	<1.2	141	<0.05	<0.1	<0.15	<0.20	2.87	<0.90	1.40	1.07	0.02	<0.01	<0.01	<0.01	0.04	1.2	2809
JC-26		39.2	48.19	83	1.61	0.03	95	7.4	200	0.44	<0.1	<0.15	<0.2	2.75	<0.9	1.11	<0.9	0.67	0.42	<0.01	<0.01	<0.01	0.6	1960
Altered		39.2	51.65	61	0.65	0.04	74	8.8	187	0.33	<0.1	<0.15	<0.2	3.57	<0.9	0.80	<0.9	0.05	0.40	<0.01	<0.01	<0.01	0.7	2096
90 %		38.5	68.67	24	0.24	0.10	242	<1.2	126	0.20	<0.1	<0.15	<0.2	1.27	<0.9	0.62	<0.9	0.33	0.32	0.04	<0.01	<0.01	2.0	3056

Table 6 (continued)

Sample	Mineral	S wt%	57Fe %	59Co ppm	61Ni %	65Cu %	66Zn ppm	75As ppm	82Se ppm	99Ru[a] ppm	103Rh[a] ppm	106Pd[a] ppm	108Pd[a] ppm	109Ag ppm	111Cd ppm	121Sb ppm	125Te ppm	185Re ppm	189Os ppm	193Ir ppm	195Pt ppm	197Au ppm	209Bi ppm	S/Se
JC-38		51	52.86	33	0.13	0.45	<40	<1.2	212	0.26	<0.1	<0.15	<0.2	3.06	<0.9	1.28	1.01	0.45	0.15	0.14	<0.01	0.02	0.5	2406
Altered	Py	51	52.52	31	0.17	<0.05	<40	<1.2	203	0.07	0.56	<0.15	<0.2	0.95	<0.9	0.90	1.38	0.23	0.11	0.21	<0.01	<0.01	0.7	2512
100 %		51	49.78	29	0.07	<0.05	<40	<1.2	194	0.12	<0.1	<0.15	<0.2	0.64	<0.9	0.75	1.08	0.46	0.11	0.15	<0.01	<0.01	0.4	2629
		38.1	61..63	7	<0.08	<0.05	<40	<1.2	<65	39.30	38.2	47.9	48.4	<0.9	<0.9	0.74	0.85	<0.02	39.3	38.2	35.4	48.5	<0.15	
Po-724 Standard		38.1	58.13	6	<0.08	<0.05	<40	<1.2	<65	39.50	37.3	46.3	47.9	<0.9	<0.9	0.79	<0.8	<0.02	37.2	37.2	34.5	49.0	<0.15	
PO-724 Standard	Certified	38.1		n/a	n/a	n/a	n/a	n/a	n/a	37.00± 1.00	37.00± 1.70	45.00± 0.80	45.00± 0.80	n/a	n/a	n/a	n/a	n/a	35.20± 1.90	36.2±0.5	35.9±0.7	47.3±2.4	n/a	

LA-ICPMS detection limits: ^{59}Co (4 ppm), ^{61}Ni (0.08 wt%), ^{65}Cu (0.05 wt%), ^{66}Zn (40 ppm), ^{75}As (1.2 ppm), ^{82}Se (65 ppm), ^{99}Ru (0.05 ppm), ^{103}Rh (0.1 ppm), ^{106}Pd (0.15 ppm), ^{108}Pd (0.2 ppm), ^{109}A(0.9 ppm), ^{111}Cd (0.9 ppm), ^{121}Sb (0.25 ppm), ^{125}Te (0.8 ppm), ^{185}Re (0.02 ppm), ^{189}Os (0.1 ppm), ^{193}Ir (0.1 ppm), ^{195}Pt (0.01 ppm), ^{197}Au (0.01 ppm), ^{209}Bi (0.15 ppm)

[a] Isotopes where corrections have been applied for polyatomic or isobaric interferences

found at the edge of silicates (Table 5 and Fig. 4). One sperrylite is located within a chromite grain (Fig. 5e and f) but most sperrylite occurs in clusters (Fig. 5g). These PGM in the silicates sometimes occur in clusters at the junctions of the altered edges of olivine and pyroxene (Fig. 5i and j). The larger sperrylite crystals identified are often euhedral and have smaller crystals of sperrylite associated with them.

Froodite is usually equant or lozenge shaped with smooth outlines. It is almost entirely hosted in magnetite. However, a few grains occur in pyrrhotite or pentlandite and rarely in chalcopyrite, sphalerite, and silicates (Table 5, Fig. 4). Commonly, BMS are altered at their edges becoming lined with selvages of magnetite. Veins of magnetite frequently crosscut-altered BMS. It is this selvage and vein magnetite that encloses the majority of the froodite grains (Fig. 6a–i).

Padmaite is usually ragged in outline and equant or irregular in shape. It is always located at the junctions of minerals either between magnetite and BMS or between different BMS (Table 5, Figs. 4 and 7a–j). In one case, a froodite grain appears to be partially altered to padmaite where it juts out into a silicate vein (Fig. 7j). Further evidence for mobility of Se is given by the presence of non-PGE-bearing Se-bearing minerals which fill veins crosscutting-altered BMS (Fig. 5h).

LA-ICP-MS

LA-ICP-MS was conducted on pentlandite, pyrrhotite, pyrite, and chalcopyrite on four samples showing a range of alteration from relatively fresh to extremely altered to determine the concentrations of PGE present in BMS (Table 6) and to compare these with the PGE whole rock assay data (Table 2, Fig. 2). The BMS are often inclusion free allowing analysis of the BMS without addition of elements from inclusions (Fig. 8).

These sulfides show a distinct lack of PGE in solid solution with individual Ru, Rh, Os, Ir, and Pt concentrations all <1 ppm and falling below detection limits in many cases. The concentrations of these PGE are similarly poor in both the fresher and the more altered samples. Pt and Ir are either very low or below detection limits in all of the sulfides analyzed despite their presence in low concentrations in the whole rock PGE analyses. What little there is of Rh, Ru, and Os tend to concentrate in pentlandite and pyrrhotite with more erratic concentrations in chalcopyrite.

In contrast, minor but appreciable Pd concentrations are consistently identified in pentlandite in both the fresh and altered samples. These measurements are much more consistent in the fresher samples (1–2.6 ppm Pd) compared to those in the more altered sample JC-26. In this sample, two Pd values of 0.72 and 0.84 ppm occur in pentlandite adjacent to chalcopyrite and three values of 2.91–3.82 and 4.20 ppm occur

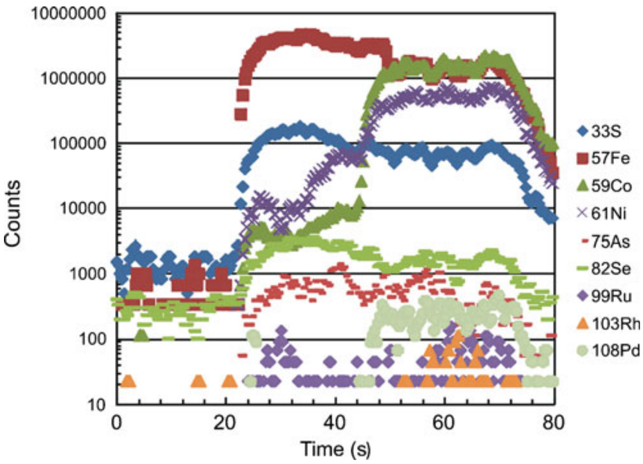

Fig. 8 LA-ICP-MS TRA spectrum for pyrrhotite and pentandite showing that these BMS are inclusion free

in pentlandite adjacent to pyrrhotite. A few erratic concentrations of Pd occur in chalcopyrite, with very low concentrations mainly below the detection limit in pyrrhotite and pyrite.

Ag and Te exhibit similar behaviors and are identified in all of the sulfides analyzed; however, concentrations of these elements are generally higher in the more altered samples. These elements also show a preference for pentlandite; a pattern which is preserved in all samples. In contrast, the semimetals As, Bi, and Sb have low concentrations throughout both the fresher and more altered samples. Average Se concentrations per sulfide phase are generally higher in the more altered samples compared to that of the fresher samples although there is considerable variation within the samples.

The majority of S/Se ratios are within the typical mantle range of 2,000–4,000 but for chalcopyrite and pentlandite S/Se ratios are consistently lower in the most altered samples and higher in the fresher samples (Table 6). Pentlandite in the most altered sample produces S/Se ratios (1,919 to 1,642) consistently lower than the mantle range.

Discussion

Magmatic concentration of PGE

The distribution of the PGE between BMS and PGM in the Jinchuan ores appears similar to that observed elsewhere. In a number of major deposits, it has now been shown that Pt does not partition into any BMS (Ballhaus and Sylvester 2000; Barnes et al. 2006; Holwell and McDonald 2007, 2010; Godel et al. 2007; Barnes et al. 2008; Godel and Barnes 2008a; Hutchinson and McDonald 2008). There is a growing body of evidence that suggests that early magmatic PGM may crystallize in association with a sulfide liquid as shown experimentally (e.g., Karup-Moller et al. 2008). The timing of PGM formation versus the incorporation of PGE into BMS appears to vary in different deposits. Even within a deposit, there is growing evidence that the presence of semimetals has an influence on the partition of PGE into the BMS especially when volatile elements, such as arsenic, are introduced into the magma from the surrounding basement (Dare et al. 2011). So Os, Ir, Ru, Rh, and Pt may crystallize to form early As-bearing PGM at high temperatures directly from the sulfide liquid, prior to the formation of MSS and ISS (e.g., McDonald 2008; Hutchinson and McDonald 2008; Dare et al. 2010b). Holwell and McDonald (2007) have shown that when Bi and Te are present, Pt, Au and some Pd will partition into a late stage semimetal-rich melt that is expelled to grain boundaries during ISS crystallization forming discrete PGM phases. Thus, the presence of different semimetals changes how different PGE behave as suggested in the cluster model originally proposed by Tredoux et al. (1995). PGE may also partition into and crystallize from an As-Bi-Sb-Te-rich melt giving rise to crystallization of As-Bi-Sb-Te PGM rather than PGE entering into solid solution in BMS (Cabri and Laflamme 1976; Tomkins, 2010; Godel et al. 2012). In the Jinchuan samples, LA-ICP-MS shows that neither Pt nor Ir are present in BMS. The irarsite grains that contain up to 12 % Pt may indicate that As scavenged the Ir and Pt to form early magmatic As-bearing PGM. The laser ablation analyses show very low concentrations of As in the fresh BMS samples suggesting that where present As is likely to be in As-rich minerals including PGM. Seven irarsites containing 11–13 % Pt were located in six polished blocks taken from the most PGE-enriched sample JZ04. However, as the PGE analysis of JZ04 contains 92 ppb Ir and over 3 ppm Pt it is unlikely that irarsite is the main host for this Pt. One sperrylite identified within a chromite crystal in the altered samples may indicate that most Pt is concentrated in sperrylite occurring as rare large crystals in JZ04. The presence of a few large Pt-bearing PGM is also suggested by the PGE assay data for the two fresher samples. This shows a very large variation in Pt concentrations (6.9–3,059 ppb) giving rise to a large variation in Pt/Pd ratios. These variations are also observed in previous studies of the Jinchuan deposit in all three ore bodies (e.g., Song et al. 2006). Song et al. (2009) suggested that these anomalies are related to magma composition whereby Pt-Fe alloys segregated prior to magma injection. This is a possibility but the evidence presented here shows that Pt-PGM may be rare but large and so the distribution of these Pt-PGM would also account for these negative and positive Pt anomalies. It is well known that Pd is usually more mobile than Pt during hydrothermal alteration (e.g., Seabrook et al. 2004) and so any subsequent hydrothermal alteration is more likely to remove Pd than Pt giving rise to positive and negative Pd anomalies rather than positive and negative Pt anomalies.

There are very few Pd-bearing PGM in the fresher Jinchuan samples suggesting that the majority is hosted in the BMS which is confirmed by the LA-ICP-MS data. Pd partitions into a Cu-rich liquid, either crystallizing to form ISS or an even more evolved late stage immiscible sulfide melt (Prichard et al. 2004; Barnes et al. 2006). However, Pd is often most concentrated in pentlandite. When the Pd/semimetal ratio is sufficiently high, excess Pd that cannot be accommodated by a late-stage semimetal-rich melt will diffuse into pentlandite (Holwell and McDonald 2010). Dare et al. (2010b) proposed that the Pd may diffuse into pentlandite from adjacent chalcopyrite during subsolidus exchange. This also may have occurred in Jinchuan where Pd is identified in solid solution, primarily in pentlandite but with minor amounts also identified in pyrrhotite and chalcopyrite. These different BMS minerals are in contact with each other in the net-textured ore and so this type of diffusion is possible.

Recent research using microbeam studies including LA-ICP-MS on PGE-bearing BMS have shown that Os, Ir, and Ru preferentially partition into pentlandite and pyrrhotite whereas Rh partitions only into pentlandite (Godel et al. 2007; Holwell and McDonald 2007). In Jinchuan, what little Os, Ru, and Rh that exists in the samples studied also tends to be concentrated into pentlandite and pyrrhotite. However, Ir values in BMS are very low suggesting that irarsite (ubiquitous throughout the fresh and altered samples, Fig. 9a) is likely to be the main host for Ir.

Alteration and sulfur loss

The debate as to whether the PGE partition into BMS or whether PGM crystallize directly from the magma is complicated by the later extensive exsolution of the PGE from the BMS during subsolidus cooling of magmatic ores. It is well known from experimental data that the solubility of PGE decreases in BMS as they cool (e.g., Makovicky et al. 1990). Thus, in many ores, BMS are accompanied by PGM formed by exsolution of the PGE during cooling. These types of PGM are usually found enclosed by or in contact with the BMS. However, PGM may also be locally remobilized away from the BMS during subsequent alteration and serpentinization as for example in the Bacuri complex in Amapa Brazil (Prichard et al. 2001) in the Shetland ophiolite (Prichard et al. 1994), in Raglan in Cape Smith (Seabrook et al. 2004), and in the Penikat intrusion (Barnes et al. 2008). The altered samples from the Jinchuan intrusion host far more PGM than the fresher samples, especially Pd-bearing PGM. This suggests that the Pd has exsolved from the BMS post magmatic alteration and subsequently crystallized as PGM, particularly as froodite phases (Fig. 9b).

Detailed study of the postmagmatic alteration of the Jinchuan intrusion by Ripley et al. (2005) found that up to 30 % of the BMS have been replaced by magnetite,

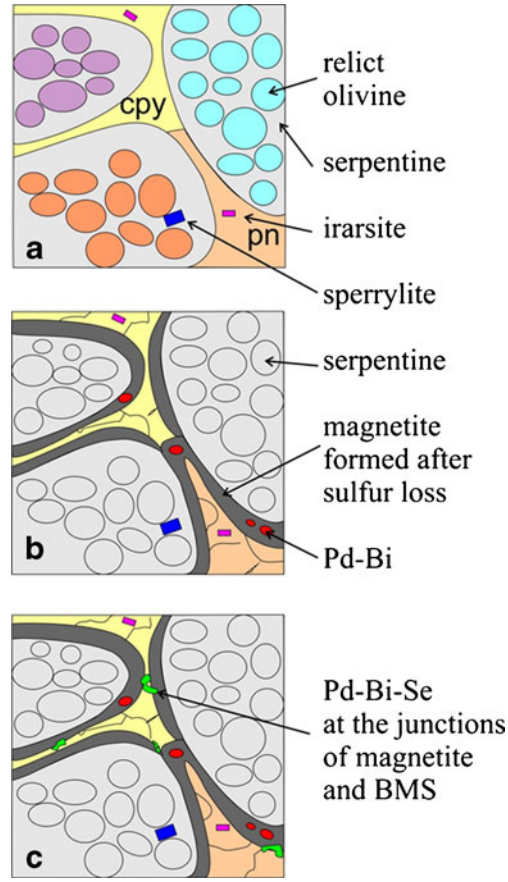

Fig. 9 Model showing three stages of alteration of the Jinchuan ores based on the samples studied from **a** fresh BMS interstitial to 70 % fresh olivine with only irarsite and sperrylite as PGM to **b** 100 % alteration of sillicates and S loss producing a magnetite selvage around altered oxidized BMS. Magnetite encloses $PdBi_2$ that joins the PGM assemblage and finally **c** with Pd-Bi-Se forming at the junctions of magnetite and BMS during late mobility of Se

serpentine, and chlorite and they also state that some of the metals and sulfur in the BMS have been removed and redeposited on a centimeter scale in fine ribbons and veins within serpentine, chlorite, and amphibole. Without sulfur loss, pentlandite alters to millerite and magnetite, pyrrhotite alters to pyrite, and chalcopyrite alters to violarite or cubanite. However, the BMS observed in this study have been affected by sulfur loss and secondary millerite, violarite and cubanite are rare with magnetite much more abundant. This confirms the findings by Ripley et al. (2005) with magnetite selvages occurring on BMS indicating sulfur loss. This is particularly significant for this study as many of the PGM identified occur within magnetite. Godel and Barnes (2008b) observe alteration of the BMS accompanied by sulfur loss and late formation of magnetite in the Stillwater complex and the resultant presence of PGM, especially Pd-bearing ones in magnetite, is similar to that observed in these Jinchuan samples.

This all supports the idea that in the Jinchuan intrusion, PGM have been formed following the release of PGE during postmagmatic alteration of the BMS that resulted in sulfur loss. This appears to be particularly true for Pd which first partitioned in pentlandite (shown by the laser ablation analyses) and was then subsequently released from the BMS during post magmatic alteration forming Pd-bearing PGM consistently associated with secondary magnetite. It is shown that pentlandite analyzed in the altered samples may both lose or retain some concentrations of Pd.

Assay data shows how whole rock Pd concentrations have remained relatively constant when compared to the fresh samples suggesting that Pd remobilization has occurred over a short distance. It appears therefore that even though alteration has released some Pd from pentlandite, it has not been significantly remobilized. Thus, froodite remains within magnetite or is still associated with BMS. Michenerite is often observed at boundaries of BMS with altered silicates or in silicate veins suggesting simultaneous transport of Pd during the mobilization of Bi and Te. Irarsite is abundant in the sheared sample JC-49 in pyrite and may have crystallized during shearing and remobilization.

Remobilization of selenium

Mobility of Se is indicated by the presence of Bi-selenide and Ag-bismuth-selenide in silicate veins in altered pentlandite (Fig. 5h). Also, the location of ragged shaped Se-bearing PGM at magnetite and BMS junctions indicates late formation of these PGM after the initial alteration of some BMS to magnetite. This is illustrated by the observation that where PGM are completely enclosed by magnetite, they have been protected from this alteration and remain Se-poor. Rarely, froodite is altered to padmaite where it is in contact with a silicate vein. Padmaite grains may form in two ways; (1) either from the alteration of froodite or (2) by precipitation directly from late stage hydrothermal fluids. Both of these processes indicate that there are two stages of alteration which is consistent with the suggestion by Ripley et al. (2005) that at least two fluids were involved in the alteration at Jinchuan. The first stage released PGE from the BMS while possibly introducing other elements such as Bi and Te forming froodite and other PGM; the second is related to the addition of Se that has resulted in the formation of Se-bearing PGM and other Se-bearing minerals. Sulfur loss during the first phase of fluid activity may be responsible for the increased concentrations of Se (and lower S/Se ratios) in the altered BMS compared with the fresh BMS while the second phase of alteration involved fluids that were capable of mobilizing Se from the remaining BMS and produced the Se-bearing PGM. The distribution of PGM-bearing selenides in the altered ores is summarized in Fig. 9c. The possible causes of this are considered below.

Distribution and conditions of formation of Se-bearing PGM Worldwide

Selenium-bearing PGM are rare and unusual. In the Jinchuan intrusion, the mineralogy, described in this paper, suggests that the Se-bearing PGM have formed as a result of hydrothermal alteration. The Se-bearing PGM occurrences worldwide are summarized in Table 7. There is a consistent set of circumstances for the transport of Se and subsequent formation of Se-bearing minerals. In the majority of cases, ore-forming fluids that transported Se are thought to be acidic and saline with high oxygen fugacity. It is also these conditions that are required to dissolve and transport PGE, principally Pt and Pd in a fluid (Wood 2002). As Pd is the most mobile PGE, it is reasonable to assume that Pd and Se are most likely to be transported and then crystallize together. Formation of Se-bearing PGM occurs with the interaction of these hydrothermal fluids carrying precious metals and Se with carbonates in the form of limestone, dolomite, or marble c.100–300 °C.

Stanley et al. (1990), Mernagh et al. (1994), and Nickel (2002) all conclude that the Pd selenides associated with carbonates formed from Se that was introduced into the system in a PGE metal-rich fluid. It is likely that fluids, of similar temperatures and characteristics to those Se-bearing fluids described elsewhere altered the Jinchuan intrusion but Se was probably sourced from the primary BMS. Extensive alteration and isotope studies (Ripley et al. 2005) shows that Jinchuan was subject to large scale hydrothermal convection involving two different postmagmatic fluids at different times and one or both of these fluids contained a significant meteoric water component. The first phase of fluid activity led to partial transformation of the BMS to magnetite and release of some PGE to form PGM. The second stage of hydrothermal alteration involved oxidized and acidic basinal fluids probably generated in the cover sequence of quartzites, conglomerates, metabasalts, shales, and limestones that form the Denzhigou Group. These oxidized brines were able to migrate downwards and altered the intrusion at temperatures of between 100 and 300 °C. During this alteration, further recrystallization of sulfides to magnetite occurred, particularly the formation of magnetite halos around existing sulfide grains as a result of sulfur loss. Conditions were sufficiently oxidizing to mobilize Se from the sulfide, possibly involving formation of selenous acid:

$$3\ FeSe_{(BMS)} + 3\ H_2O + 5\ O_2 = Fe_3O_{4(magnetite)} + 3\ SeO_3{}^{2-}{}_{(aq)} + 6\ H^+{}_{(aq)}$$

Table 7 Occurrances of Se-bearing PGM worldwide

Locality	PGM	Host lithology	Temperature of formation	Reference
Hope's Nose, Devon UK	Chrisstanleyite ($Ag_2Pd_3Se_4$)	Au-bearing carbonate veins in Middle Devonian limestone	Epithermal saline fluid, high fO_2 low pH, 120 °C	Paar et al. (1998), Stanley et al. (1990)
Coronation Hill, Australia	Palladseite ($Pd_{17}Se_{15}$) milotaite (PdSbSe)	Ferroan dolomite vein in carbonaceous tuff	Ca-rich acidic brine, high fO_2 Pd + Se from cover descending to basement at 140 °C.	Mernagh et al. (1994)
Copper Hills, Australia	Oosterboschite $(Pd,Cu)_7Se$ luberoite (Pt_5Se_4)	Veins of quartz and malachite in dolomitic carbonate	Acid saline fluid carrying precious metals and Se	Nickel (2002)
El Chire, Argentina	Jaguéite ($Cu_2Pd_3Se_4$) chrisstanleyite	Calcite-bearing vein	High fO_2	Paar et al. (2004)
Musonoï Mine, DRC	Verbeekite ($PdSe_2$) oosterboschite	Quartz and dolomite	Se-U-PGE-rich fluid>200 °C	Pirard and Hatert (2008)
Harz Mts, Germany	Tischendorfite ($Pd_8Hg_3Se_9$)	Calcite and ankerite veins in black shale	High fO_2 Pd + Se originating from black shale 100 °C	Stanley et al. (2002)
Předbořice Czech Rep.	Milotaite	Carbonate veins in metamorphic rocks	Not discussed	Paar et al. (2005)
Serra Pelada, Brazil	Sudovikovite ($PtSe_2$), palladseite , Pd-Pt-Se alloy, Pd-Pt-Se, Pd-Se, Pd-Hg-Se, Pd-Bi-Se	Brecciated carbonaceous rocks between dolomitic marble and carbonaceous meta-siltstone	Not discussed	Cabral et al. (2002a, b)
Cavalcante Brazil	Kalungaite (PdAsSe) padmaite (PdBiSe)	Quartz-muscovite mylonite	Acidic and saline fluid of peraluminous granitic origin interacted with country rock	Botelho et al. (2006)
Gongo Soco MG Brazil	Chrisstanleyite $Pd_8Hg_3Se_9$, $Pd_5(Hg,Sb,Ag)_2Se_6$ $(Pd,Sb,Ag,Hg)_5Se_4$	Hematitic veins in Itabira iron fm	Saline, high fO_2 hydrothermal below 150 °C.	Cabral and Lehmann (2007)
Itabira MG Brazil	Sudovikovite			
	palladseite	Hematite and chromiferous magnetite	Saline, high fO_2 up to 300 °C.	Cabral and Lehmann (2007); Kwitko et al. (2002)
Cauê MG Brazil	Jacutingaite (Pt_2HgSe_3)	Fe ore	Not discussed	Vymazalová et al. (2011)

Where the Se-bearing fluid was able to interact with Bi-bearing PGM, namely froodite, it could react to produce padmaite as follows:

$$6\ PdBi_{2(froodite)} + 6\ SeO_3{}^{2+}{}_{(aq)} + 18\ H^+{}_{(aq)}$$
$$= 6\ PdBiSe_{(padmaite)} + \left[Bi_6(OH)_{12}\right]^{6+} + 3\ H_2O$$
$$+ 1.5\ O_2$$

The Bi was removed in the form the of the $[Bi_6(OH)_{12}]^{6+}$ complex which is soluble up to near-neutral pH (Cotton and Wilkinson 1980). This insensitivity to pH would allow Bi to be widely mobilized before it precipitated out as bismithide minerals. Migration of selenous acid with low pH fluid allows mobility of Se. Pd is also mobilized by oxidizing low pH fluids (Wood 2002). When these fluids encounter reactive host rocks such as carbonates, which neutralize the low pH solution, Se is deposited in the form of selenide minerals which are sometimes Pd-bearing.

Implications for S/Se ratios

S/Se ratios have been used to determine the source of sulfur in magmatic deposits, such as whether it is derived from the mantle or is introduced by contamination and assimilation of crustal material to achieve sulfur saturation (Eckstrand et al. 1989). For example, the Merensky Reef in the Bushveld complex has low S/Se ratios indicating a mantle S origin (Barnes and Maier 2002) whereas high S/Se ratios in the norite of the Duluth Complex suggest that sulfur was introduced as a result of crustal contamination (Thériault and Barnes 1998). Use of high S/Se ratios coupled with $\delta^{34}S$ that is significantly deviant from the mantle range ($\delta^{34}S$=0±3 ‰) can give double confirmatory evidence of S contamination of a magma (Queffurus and Barnes 2010) and is more reliable than using S/Se ratios alone. In addition, S/Se ratios are often used to identify any loss of sulfur from the system during alteration (Godel and Barnes 2008a). Selenium readily substitutes for S in many BMS and has been thought to be relatively immobile in comparison with sulfur which is more mobile and therefore more easily lost during hydrothermal activity or through reaction between sulfide and Cr spinel. High PGE tenors compared to a lower than expected sulfide content is often explained by sulfur loss (Naldrett and Lehmann 1988).

The mobility of Se and formation of Se-bearing PGM as described in this study and elsewhere, as reviewed here, has implications for the use of S/Se ratios in hydrothermally altered complexes. Selenium is usually uniformly distributed among pyrrhotite, pentlandite, and chalcopyrite (Paktunc et al. 1990; Czamanske et al. 1992). The LA-ICP-MS analyses obtained during this work confirm this (Table 6) and also show that S/Se ratios of BMS at Jinchuan are significantly lower in the altered samples compared with the fresh samples; a feature probably related to the earlier of the two fluid events recognized by Ripley et al. (2005). Also in Jinchuan, Se has been remobilized by a later, highly oxidized fluid to form Se-bearing PGM. The evidence presented here shows that Se can be remobilized over both small and much greater distances during oxidizing post magmatic low temperature hydrothermal alteration.

Therefore, if Se is likely to have been remobilized, especially where an igneous complex is subject to alteration from oxidized basinal fluids, then care needs to be taken in the use of S/Se ratios to calculate the original sulfur content of magma. The host minerals for the Se need to be identified because whole rock S/Se ratios may differ from ratios of S/Se in BMS if some of the Se has been remobilized locally to form Se-bearing PGM. In addition, there may have been more extensive remobilization of the Se into veins hosted in reactive country rocks such as carbonates some distance from BMS that provided the source of the Se. Loss of Se as well as S will give an apparently lower sulfur loss and a resulting higher S/Se ratio. Thus, if Se is mobile and is removed, this will distort the estimation of S loss using S/Se calculations. In addition, estimates of the S introduced during magma contamination will be high due to higher S/Se ratios produced if Se has been remobilized and lost subsequently during oxidizing alteration conditions. Evidence from Jinchuan and other Se-bearing PGM occurrences show that Se cannot always be assumed to be immobile when considering S/Se ratios.

Conclusions

This study of PGE in the Ni-Cu sulfide ores in the Jinchuan intrusion has shown that during the initial magmatic concentration of the PGE Pd partitioned into BMS whereas the presence of arsenic may have been responsible for the early crystallization of irarsite and possibly sperrylite resulting in a deficit of Ir and Pt in the BMS. During an early stage of hydrothermal alteration, the PGE, particularly Pd, have been released from the BMS during sulfur loss resulting in the formation of Pd-PGM. A second phase of alteration involving oxidized fluids at low temperatures mobilized Se from the BMS to form padmaite (PdBiSe). Finally, the mobility of Se, if unaccounted for, is detrimental to the study of S/Se ratios. Therefore, care should be taken when S/Se ratios are used to demonstrate sulfur mobility as it is clear from this study and our review of Se-bearing PGM, that reactions involving oxidized fluids can remobilize Se. The presence of Pd selenides located in carbonates worldwide suggests that fluids precipitate Pd and Se on interaction with carbonates and both Pd and Se may be sourced from magmatic PGE-bearing BMS ores. This paper describes an initial study of PGE distribution in Jinchuan ores and further research is necessary to establish differences of PGE distribution in different types of ores and in different ore bodies.

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